1 Tensors

Definition 1.1 (Tensors)**.**
If $\dim \mathcal{M} \neq \infty$, the tensors of type (r, s) $T_p^{r,s}\mathcal{M}$ are all the linear functions

$$f : \bigtimes^{r} T_p^*\mathcal{M} \times \bigtimes^{s} T_p\mathcal{M} \to \mathbb{R}.$$

I.e., it eats r covectors and s vectors.

Theorem 1.1 (Dimensions of General Tensor Space)**.**
The dimension of $T_p^{r,s}\mathcal{M}$ is $m^r m^s$. In particular, a basis for the space is,

$$\bigotimes_{1\leq\mu_1\cdots\mu_r\leq m} \left(\partial_{\mu_i}\right)_p \otimes \bigotimes_{1\leq\nu_1\cdots\nu_s\leq m} (dx^{\nu_i})_p$$

Remark.
For a detailed proof, see Hoffman. ♍

Theorem 1.2 (Transformation Properties of Tensor Fields)**.**
Given a manifold $\mathcal{M}$ of dimension m, choose two overlapping charts $\phi : U \to V$ with local coordinates $x^1, \dots, x^m$, $\phi' : U' \to V'$ with ${x'}^1, \dots, {x'}^m$. A tensor field T of type (r, s) with local representation on U given by

$$T^{\mu_1\cdots\mu_r}{}_{\nu_1\dots\nu_s}\,\partial_{\mu_1} \otimes \cdots \otimes \partial_{\mu_r} \otimes dx^{\nu_1} \otimes \cdots \otimes dx^{\nu_s}$$

transforms like

$$T^{\mu_1'\cdots\mu_r'}{}_{\nu_1'\cdots\nu_s'} = T^{\mu_1\cdots\mu_r}{}_{\nu_1\dots\nu_s}\frac{\partial {x'}^{\mu_1'}}{\partial x^{\mu_1}} \cdots \frac{\partial {x'}^{\mu_r'}}{\partial x^{\mu_r}} \frac{\partial x^{\nu_1}}{\partial {x'}^{\nu_1'}} \cdots \frac{\partial x^{\nu_1}}{\partial {x'}^{\nu_s'}}$$

Definition 1.2 (Pullback of Covariant Tensor)**.**
The pullback of a general tensor field τ by $f : \mathcal{M} \to \mathcal{N}$ is defined as

$$(f^*\tau)(X_1, \dots, X_k) := (\tau \circ f)(f_*X_1, \dots, f_*X_k).$$

1.1 The Metric Tensor

1.1.1 Riemannian Metric

Definition 1.3 (Riemannian Metric)**.**
A Riemannian metric on a manifold $\mathcal{M}$ is a totally symmetric, smooth $(2,0)$-tensor field on $\mathcal{M}$ that is positive definite at every point $p \in \mathcal{M}$. Locally,

$$g = g_{ij} dx^i \otimes dx^j.$$

Definition 1.4 (The Euclidean Metric)**.**
On the Euclidean space $\mathbb{R}^m$, there is a natural metric called the Euclidean metric, which is locally defined as

$$\bar{g} = \delta_{ij} dx^i \otimes dx^j.$$

Definition 1.5 (The Product Metric)**.**
Given two Riemannian manifolds $(\mathcal{M}, g_{\mathcal{M}}), (\mathcal{N}, g_{\mathcal{N}})$, we can form a metric $\hat{g}$ on the product manifold $\mathcal{M} \times \mathcal{N}$ by

$$\hat{g}((v_1, v_2), (w_1, w_2)) = g_{\mathcal{M}}(v_1, w_1) + g_{\mathcal{N}}(v_2, w_2).$$

1.1.2 Existence of Riemannian Metric

Theorem 1.3 (Existence of Riemannian Metric)**.**
Every smooth manifold admits a Riemannian metric Definition 1.3.

Proof. Let the manifold have coordinate charts $\phi_j, j \in J$. Let a smooth partition of unity $\rho_j, j \in J$ subordinate to U_j. Then we can form a smooth tensor field

$$g = \sum_j \rho_j \phi_j^* g_j,$$

where g_j is the Euclidean metric on V_j. It is obviously symmetric. Since the partition of unity is positive and every g_j is positive definite, g is positive definite also. ▨

1.1.3 Pullback Metric

Definition 1.6 (The Pullback Metric)**.**
Let $f : \mathcal{M} \to \mathcal{N}$ be C^∞, where $\mathcal{N}$ is a Riemannian manifold equipped with Riemannian metric g. Then f^*g is a Riemannian metric on $\mathcal{M}$ iff f is a smooth immersion.

Theorem 1.4 (A Useful Mnemonic for the Pullback Metric)**.**
Let $f : \mathcal{M} \to \mathcal{N}$ be a smooth immersion so that Definition 1.6 can be defined. Then

$$(f^*g) = (g_{ij} \circ f)d(y_i \circ f)^2 \otimes d(y_j \circ f)^2,$$

where $d(y_i \circ f)$ expands like $\frac{\partial(y_i \circ f)}{\partial x^j}dx^j$, just like the exterior derivative.

Proof.

$$\begin{aligned} f^*g(X_1, X_2) &= (g \circ f)(f_*X_1, f_*X_2) \\ &= (g_{ij} \circ f)dy^i(f_*X_1)dy^j(f_*X_2) \\ &= (g_{ij} \circ f)dy^i(\partial_{\nu_1}\frac{\partial f^{\nu_1}}{\partial x^{\mu_1}}X_1^{\mu_1})dy^j(\partial_{\nu_2}\frac{\partial f^{\nu_2}}{\partial x^{\mu_2}}X_2^{\mu_2}) \\ &= (g_{ij} \circ f)\frac{\partial f^i}{\partial x^{\mu_1}}\frac{\partial f^j}{\partial x^{\mu_2}}X_1^{\mu_1}X_2^{\mu_2}. \end{aligned}$$

So

$$(f^*g)_{\mu\nu} = (g_{ij} \circ f)\frac{\partial f^i}{\partial x^{\mu_1}}\frac{\partial f^j}{\partial x^{\mu_2}}.$$

But

$$d(y^i \circ f) := \frac{\partial(y^i \circ f)}{\partial x^j}dx^j.$$

Tensor product all those completes the proof. ▨

1.1.4 Musical Isomorphism

Definition 1.7 (Musical Isomorphism)**.**

$$\begin{aligned} {X^\flat}_\mu &= g_{\mu\nu}X^\nu \\ {\omega^\sharp}^\mu &= g^{\mu\nu}\omega_\nu, \end{aligned}$$

where $g^{\mu\nu} := g^{-1}_{\mu\nu}$.